

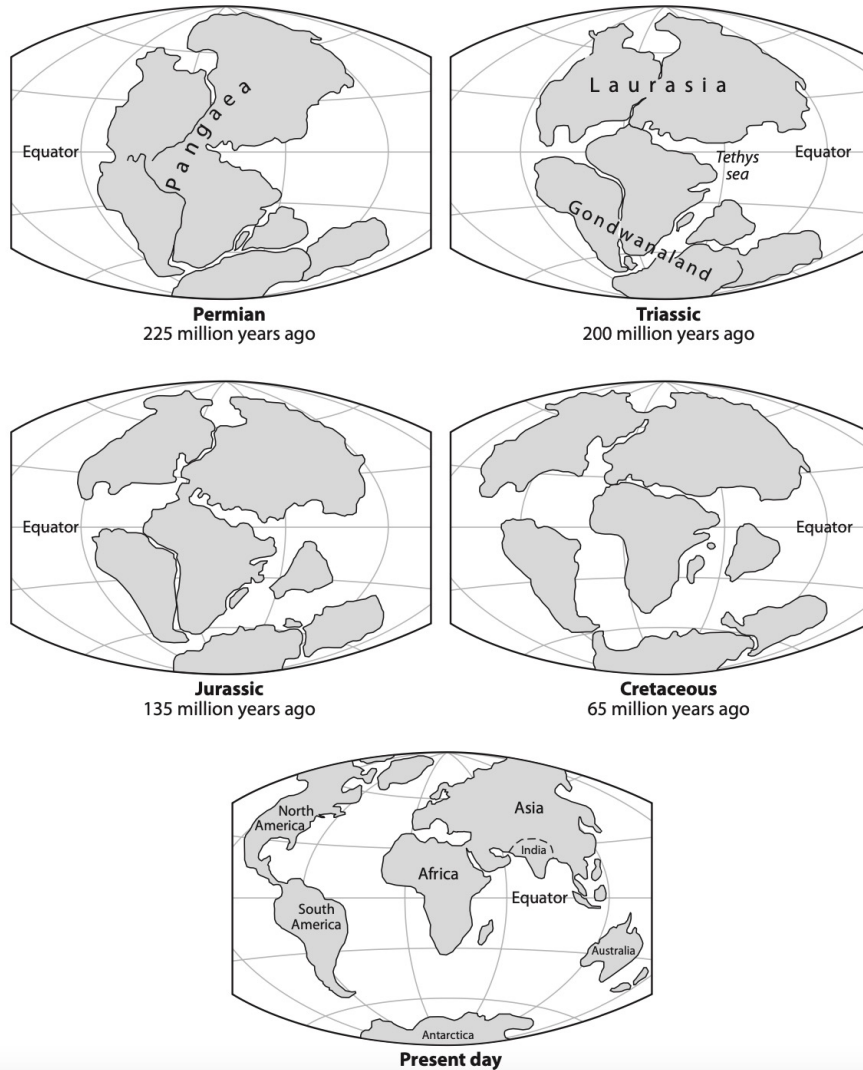
8. Planetary Formation

From dust to planets

2025 November 3

1. Introduction

200 Myr ago



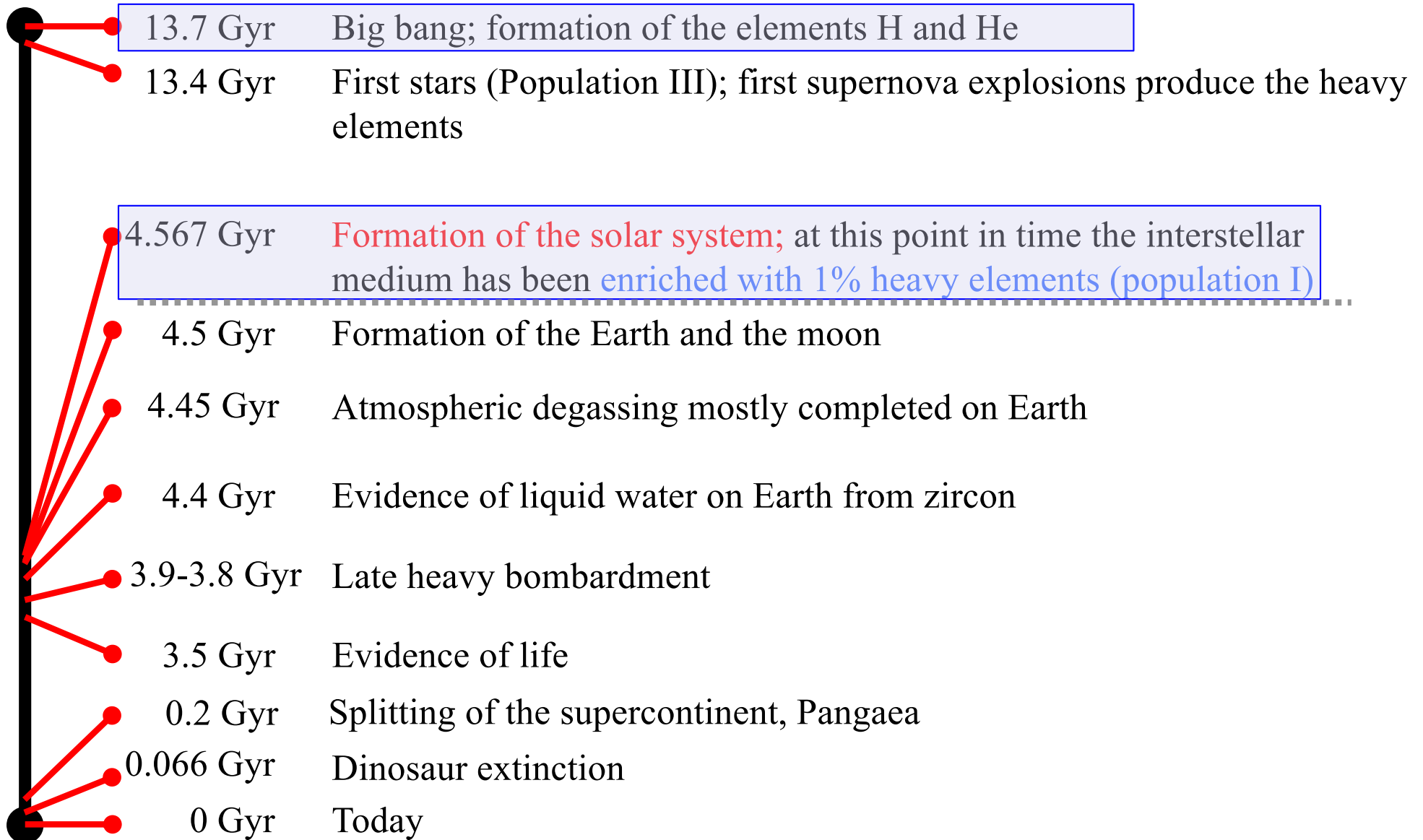
66 Myr ago



Fig. 17-3: The Cretaceous-Tertiary extinction from La Sierrita, showing the light gray claystone that marks the boundary. Notice the very brief stratigraphic interval over which the boundary occurs. (Photo courtesy of Gerta Keller, Princeton University)

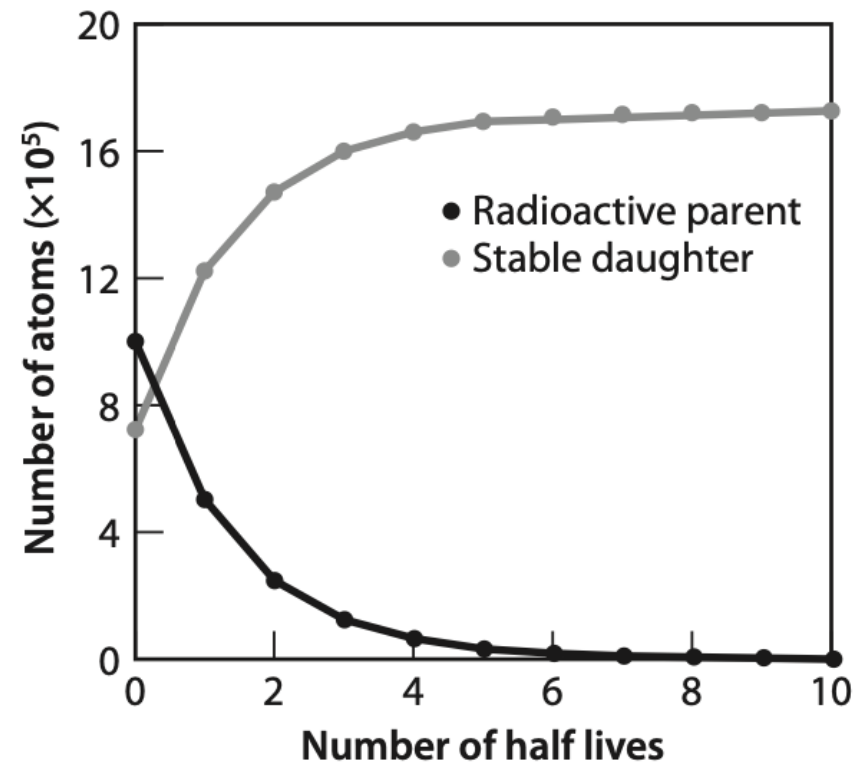
→ From millions (Myr) to billions (Gyr)

2. Chronology of the Universe



3. Measuring the Ages

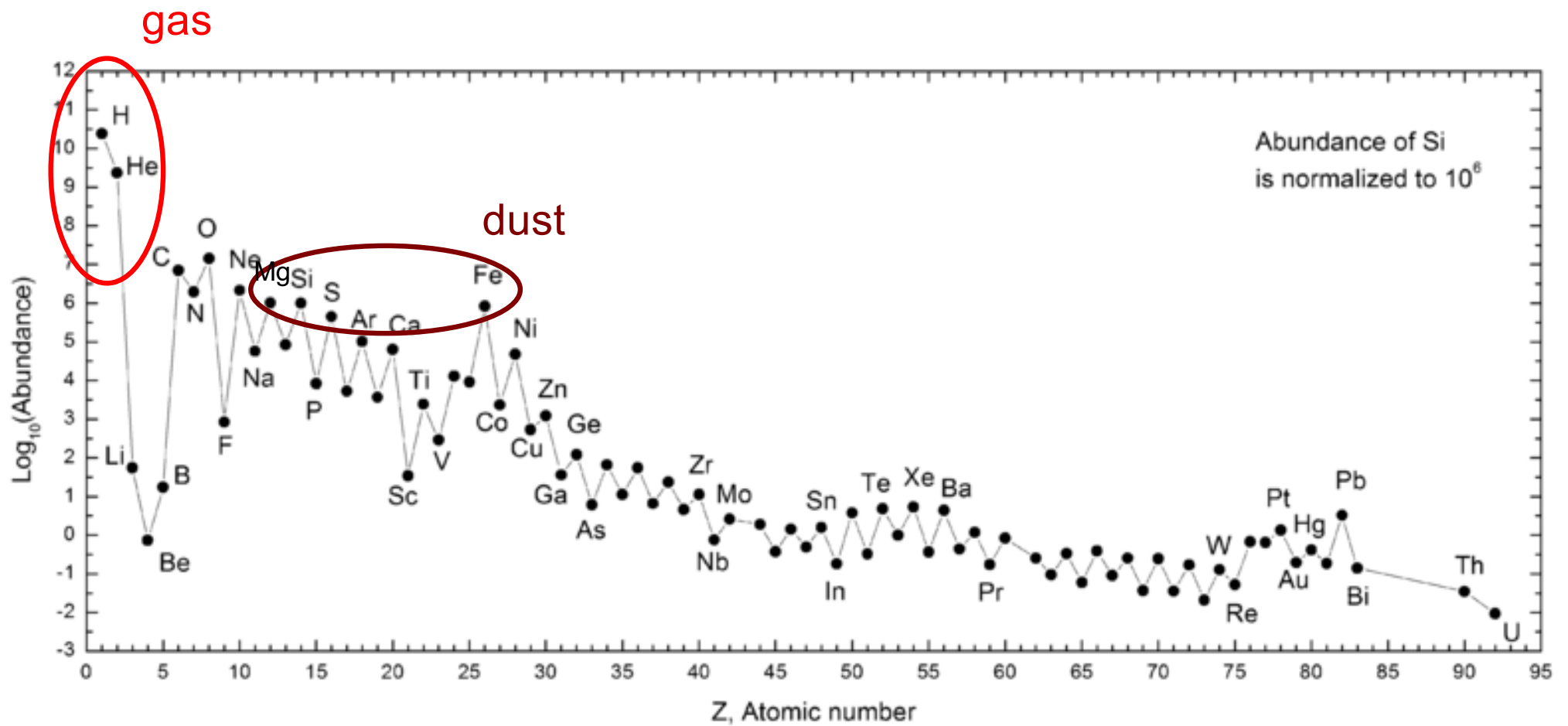
- Some isotopes are unstable → they decay into stable ones.
→ The time scale is called the half-life.
- Example (short-lived):
 - ^{14}C → used for dating archaeological samples (half-life = 5,730 yr)
- Example (long-lived):
 - ^{87}Rb → ^{87}Sr (half-life = 48.8 Gyr)
 - Produced early in the Solar System (likely by a supernova)
- The ratio $^{87}\text{Sr} / ^{86}\text{Sr}$ decreases over time
 - used to estimate the age of rocks and meteorites



4. Ingredients of planetary systems (1)

- Meteorites and Planets -

- Primitive meteorites → key to estimate the original materials of the Solar System.
- Giant planets: difficult to study — covered by thick atmospheres.
- Terrestrial planets (e.g., Earth): difficult to study
 - Samples are available.
 - But composition is biased —
 - Light elements (H, He) escape to space.
 - Heavy metals (e.g., Fe) sink deep into the core.
- **Primitive meteorites = survivors (“fossils” of the early Solar System)**
 - Preserve the original composition of the solar nebula.
- Note: Few light elements (H, He) remain in meteorites.



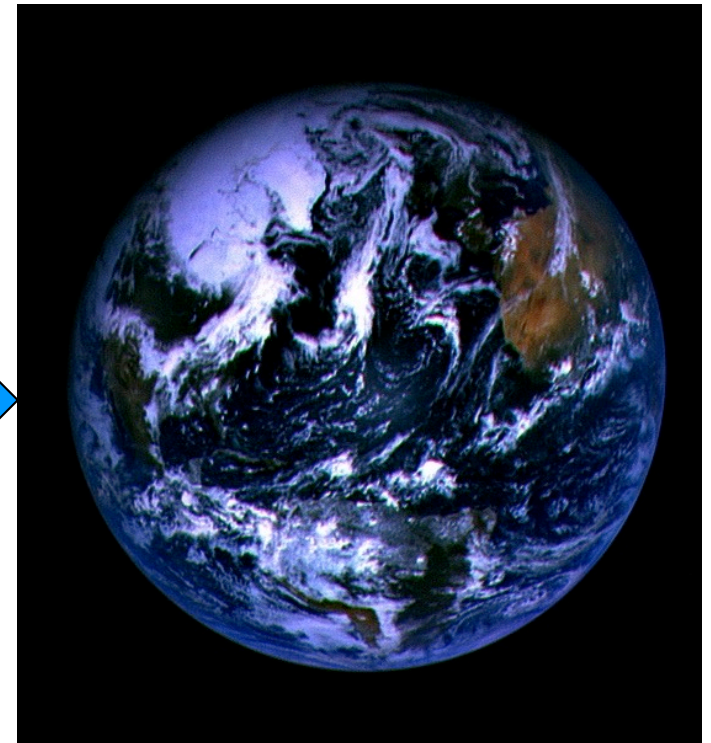
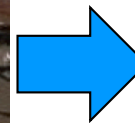
Lesson from the Sun and primitive meteorites

4. Ingredients of planetary systems (2)

- From Elements to Molecules -

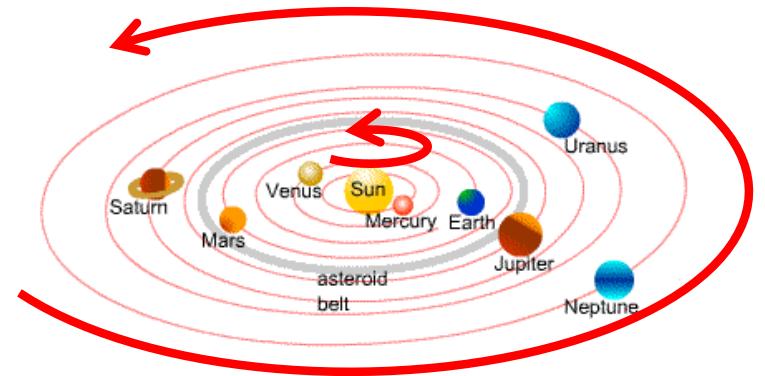
- **H** → mainly as gas H_2 , partly as ice (CH_4 , NH_3 , H_2O)
- **He, Ne, Ar** → noble gases, exist only as gas
- **C, N** → mainly in ices (CH_4 , NH_3 , CO , CO_2)
- **O** → very reactive
 - combines with H and C → ice
 - combines with metals → oxides (solid)
- **Mg, Al, Si, Fe** → metals that form solid oxides
- **S** → behaves like O
 - forms gas H_2S , or solid FeS
- Molecules exist as gas or solid (depending on temperature).
 - In this class, solid particles = “dust.”

- “We can think about planet formation like this:
 - Many a little makes a mickle.
 - Light gains make heavy purses.
 - Little drops of water make the mighty ocean
 - 티끌 모아 태산
- Many small things come together to make something big.
- Just like small coins make a large amount of money, tiny dust and gas particles eventually form planets.



5. Simultaneous Formation of a Star and Planets

- **The Sun and planets formed at the same time** and from the same material reservoir, based on the following evidence:
 1. Similar elemental abundances
 2. Similar ages
 3. In addition, the angular momentum of the Sun is aligned with the planetary orbital planes



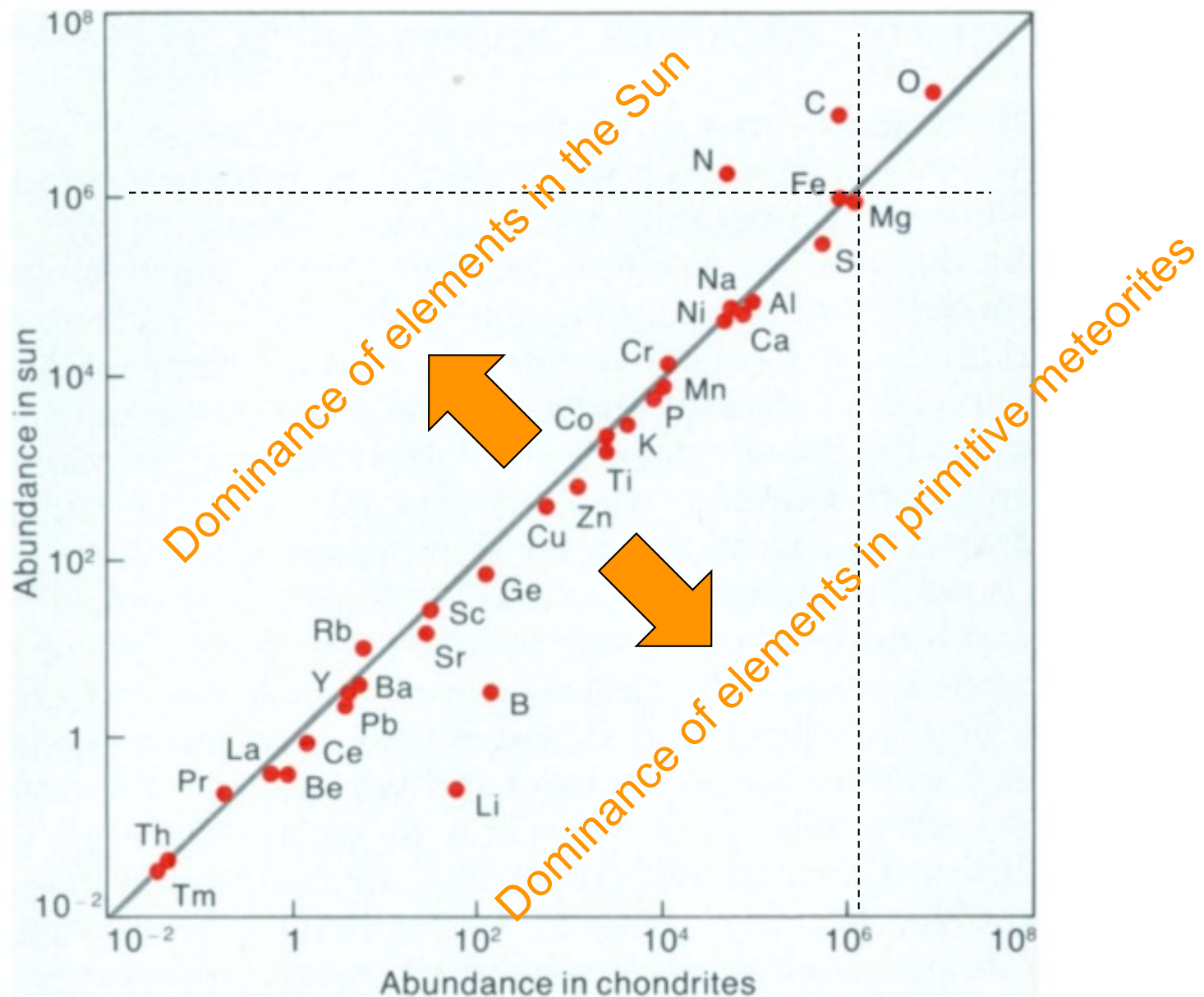
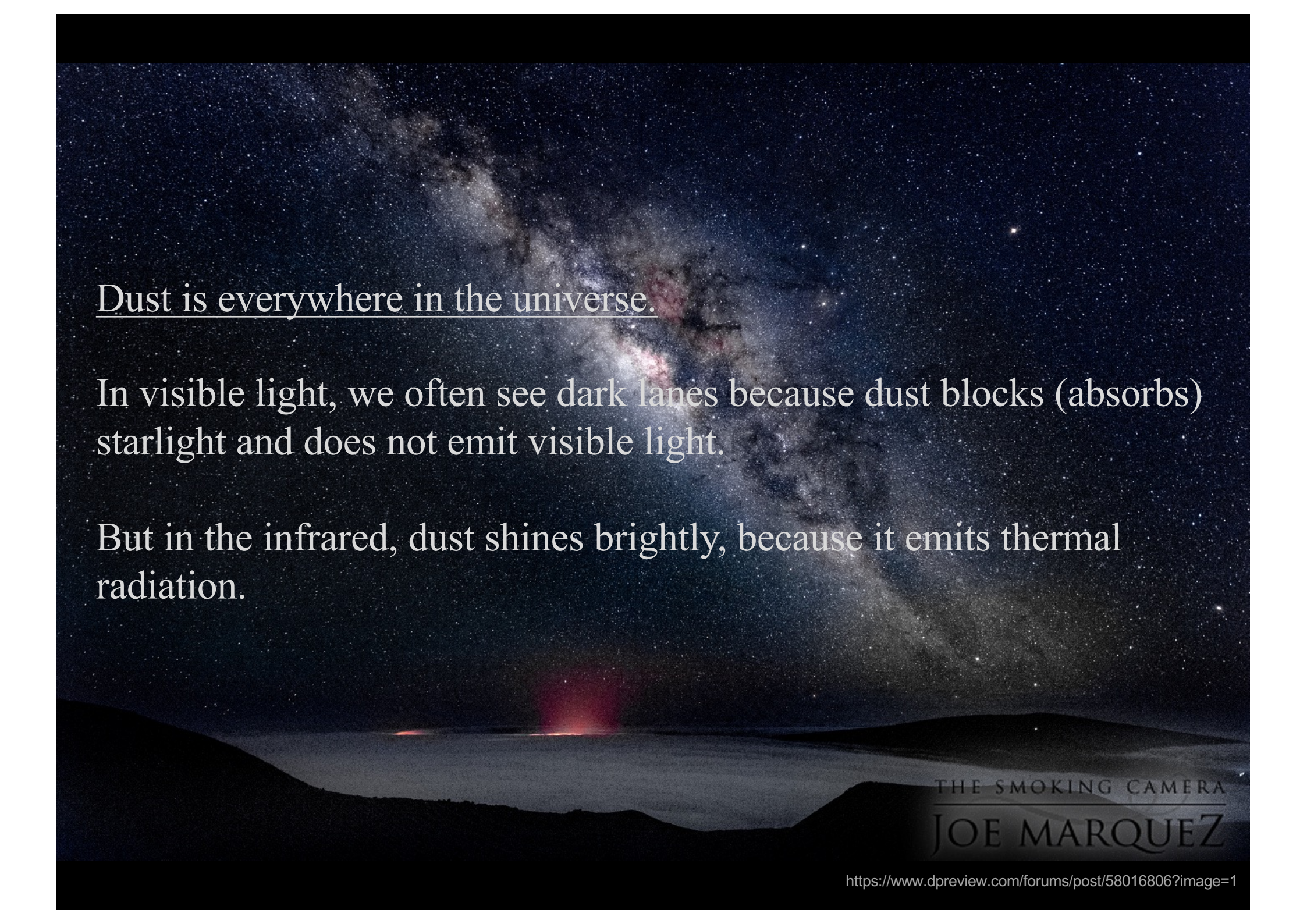


Figure 4. The chemical composition of chondrites closely matches the composition of the sun, indicating that the chondrites have not been altered chemically by geologic processes. This graph compares the atomic abundances of 33 elements, relative to 10^6 atoms of silicon. The scale is logarithmic.



THE SMOKING CAMERA
JOE MARQUEZ

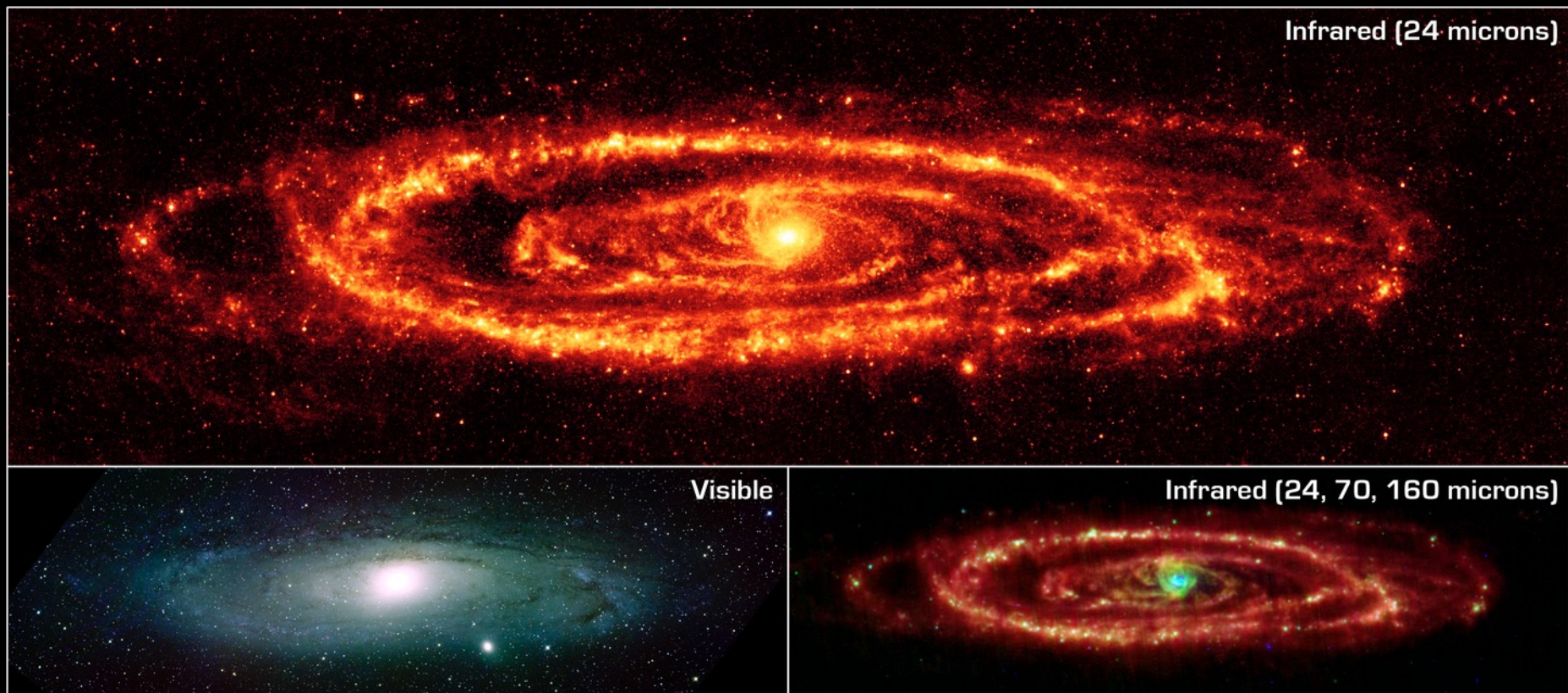


Dust is everywhere in the universe.

In visible light, we often see dark lanes because dust blocks (absorbs) starlight and does not emit visible light.

But in the infrared, dust shines brightly, because it emits thermal radiation.

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Dust in Andromeda Galaxy (M31)

NASA / JPL-Caltech / K. Gordon (University of Arizona)

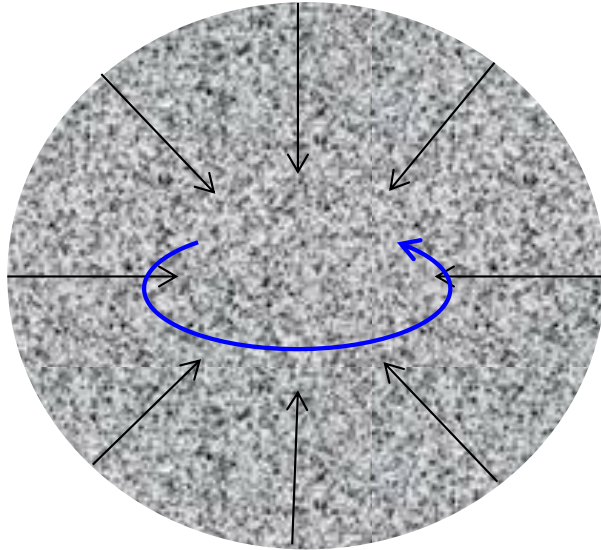
Spitzer Space Telescope • MIPS

Visible: NOAO/AURA/NSF

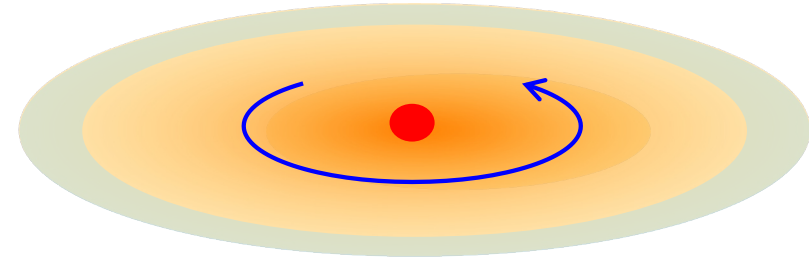
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The top image shows the Andromeda Galaxy (M31) in infrared, where you can clearly see bright rings of dust. In visible light (bottom left), however, the same regions appear dark or obscured.

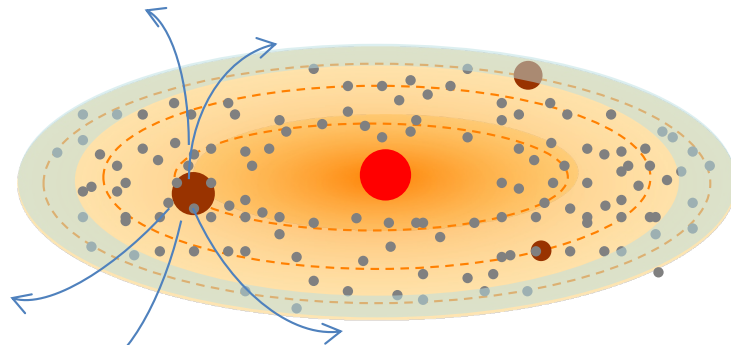
7. Formation of a star and planets



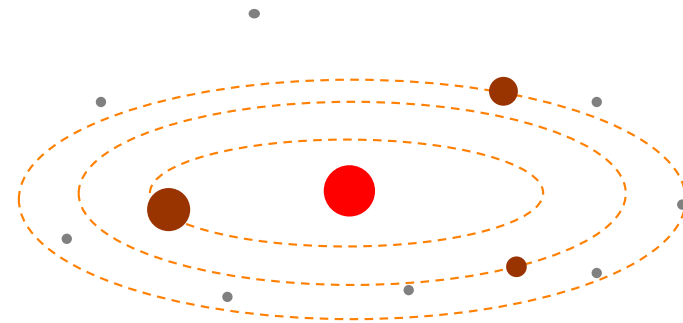
- A molecular cloud collapses
→ Temperature rises, rotation increases.



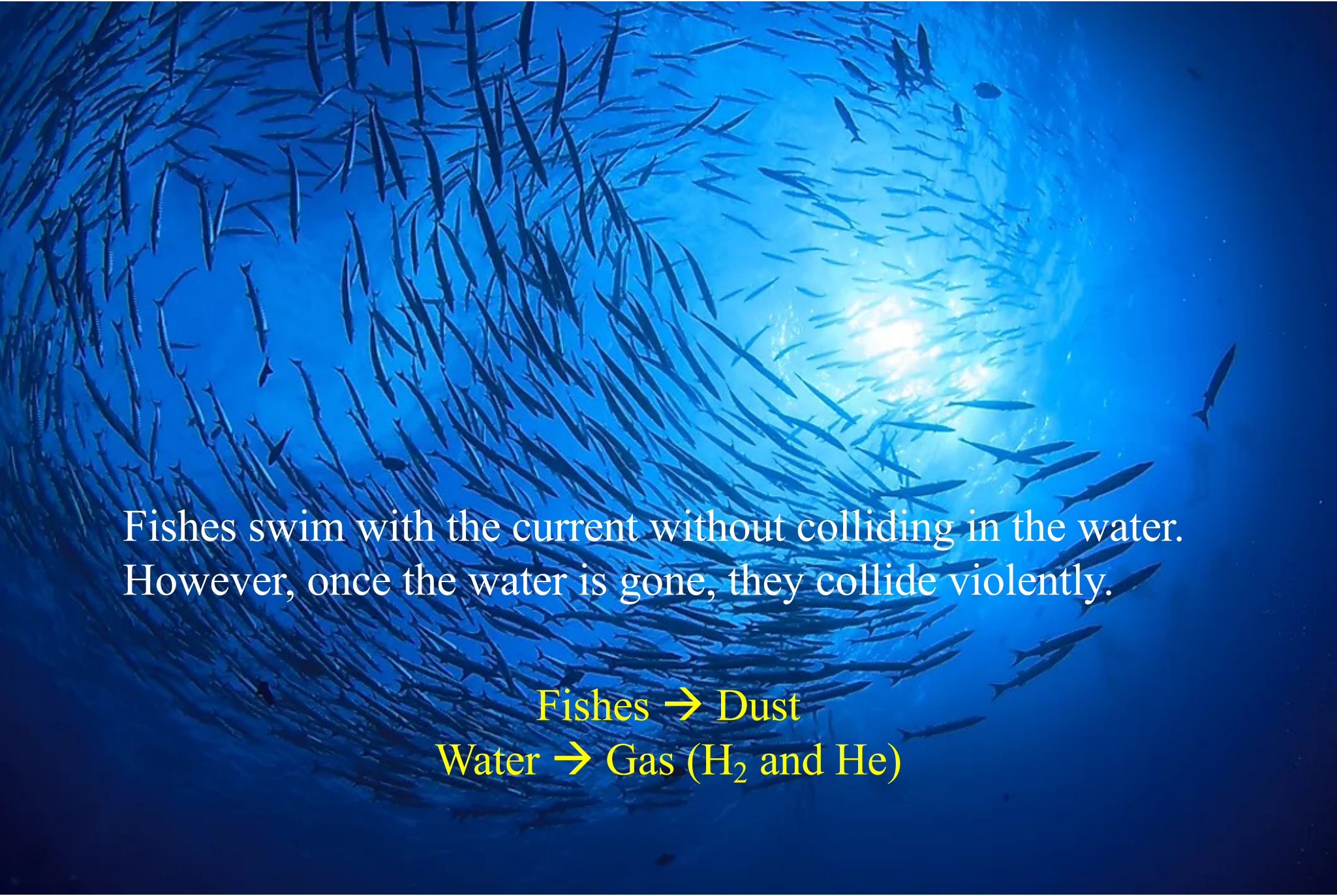
- Angular momentum is conserved, and a rotating disk forms
→ Gas and dust rotate together.



- Dust grains stick by electrostatic forces
→ form planetesimals (km-size).
→ Later, gravity dominates → proto-planets grow.

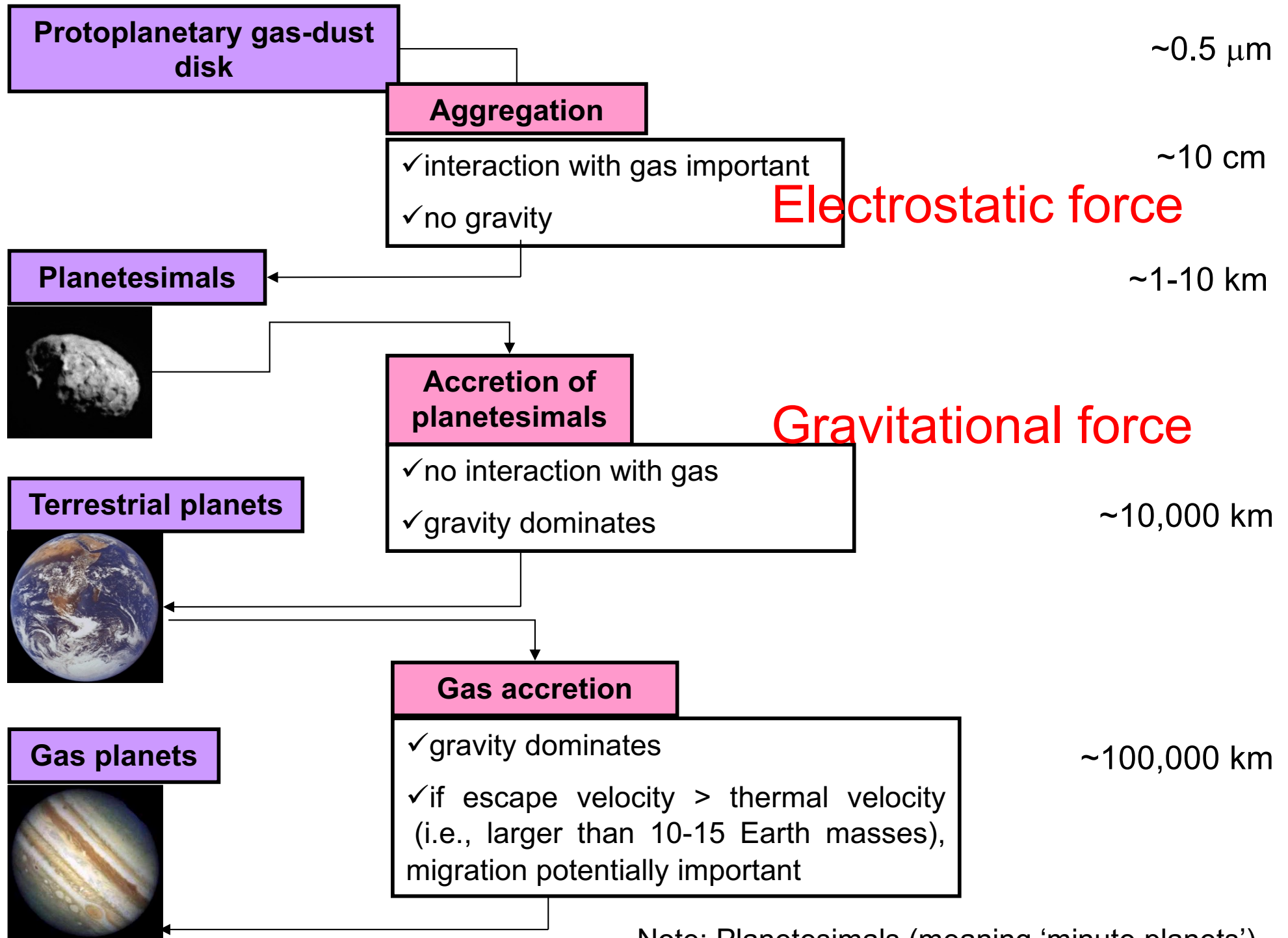


- The Sun starts fusion, and stellar wind blows away gas (H_2 , He).
→ After gas loss, planetary growth stops.

A large school of fish swimming in the ocean, illustrating the concept of fluid flow. The fish are densely packed and moving in a coordinated manner, creating a sense of motion and flow. The water is a deep blue, and the sunlight filters through from above, creating a bright, glowing effect in the center of the school.

Fishes swim with the current without colliding in the water.
However, once the water is gone, they collide violently.

Fishes → Dust
Water → Gas (H_2 and He)



8. Evidence for planetary formation



Rosetta Mission (ESA)

A comet = A possible planetesimal

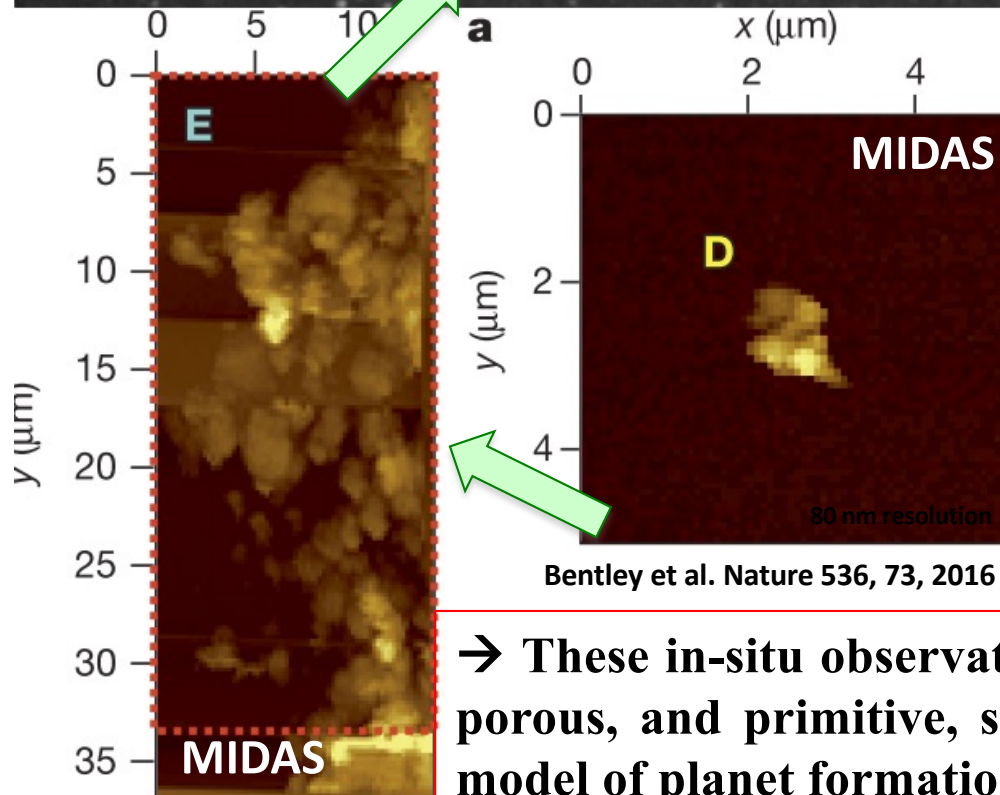
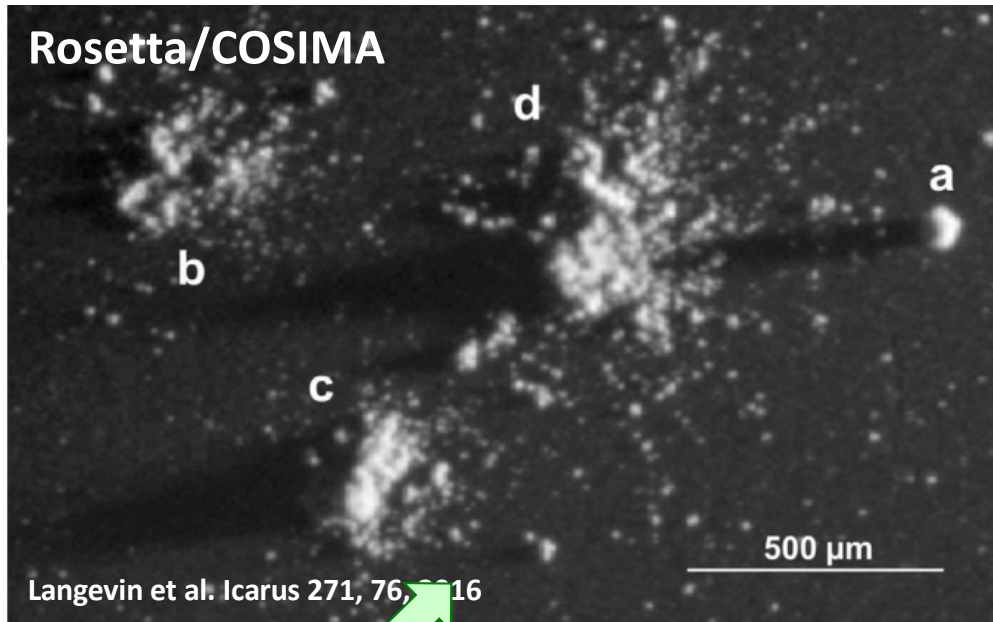
8. Evidence for planetary formation

- “Until recently, this planetary formation theory was mainly based on computer simulations and theoretical models.
- However, in the last decade, direct observational evidence has finally been obtained through space missions.
- One of the most remarkable examples is the Rosetta Mission by the European Space Agency (ESA).
- Rosetta explored Comet 67P/Churyumov–Gerasimenko, a remnant of a possible planetesimal.

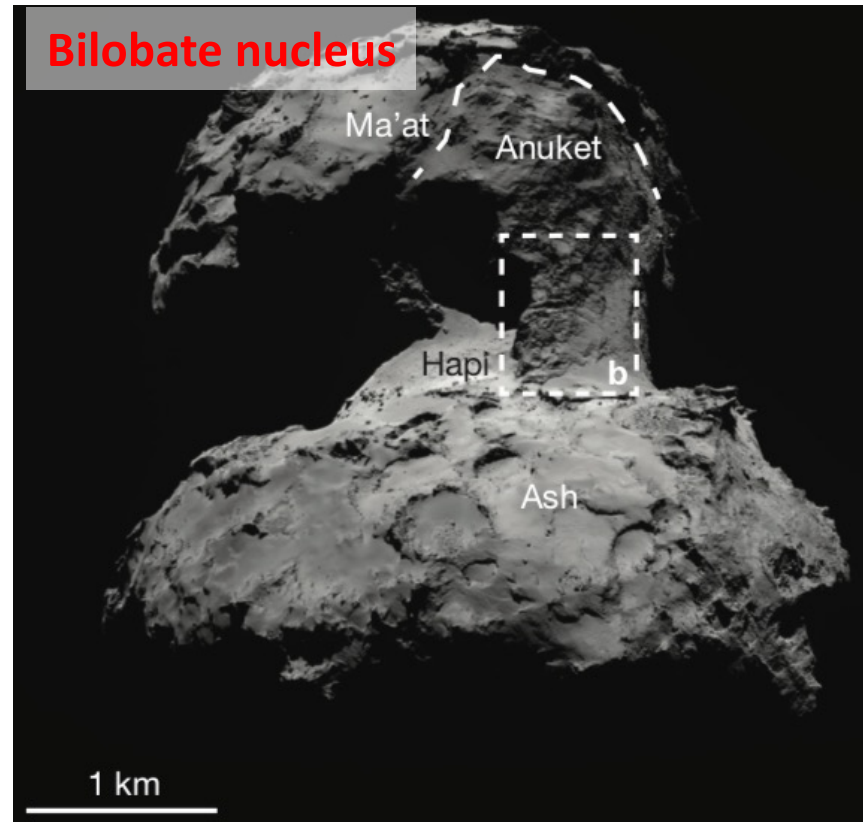


Comet 67P/Churyumov-Gerasimenko by Rosetta mission

Rosetta/COSIMA



Bilobate nucleus



Massironi et al. Nature 402, 526, 2015

→ These in-situ observations revealed that cometary material is fluffy, porous, and primitive, strongly supporting the hierarchical accretion model of planet formation (Davidsson+2016).

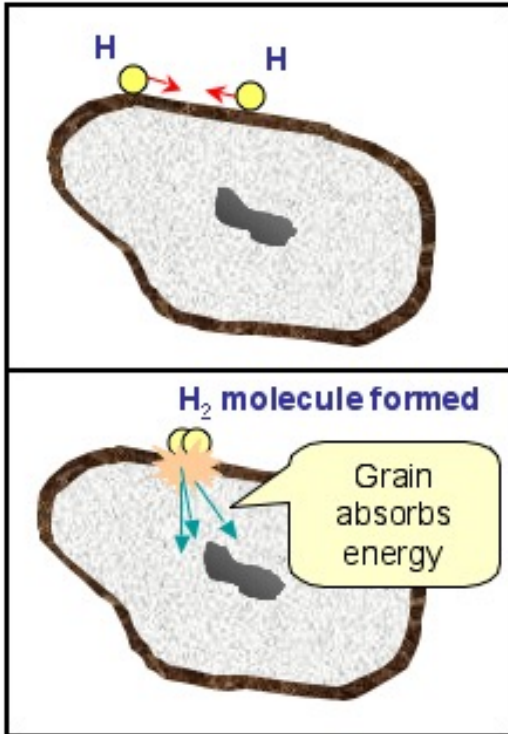
9. What have we learned so far?

- The Sun and the planets formed together about 4.567 Gyr ago.
- Dust plays a crucial role in the formation of planets.
- The lifetime of protoplanetary disks, the birthplaces of planets, is only a few million years.
- Extrasolar planetary systems can be similar to, or different from, our solar system, depending on the initial conditions during the star and planet's formation.

Summary: Star and Planetary Formation with Organic Molecules

1. Let me summarize the key points about stars and planetary formation.
2. Hydrogen and helium are distributed throughout the universe.
→ Under sufficient temperature and pressure, they undergo nuclear fusion → Stars
3. Through these reactions, heavier elements are synthesized, and energy is released
→ Preparation for dust.
4. The birth and death of stars produce materials essential for habitable planets —
→ dust and molecules such as water and organics. (What about proteins?)
5. Dust particles act as chemical factories for molecular reactions.
→ On their surfaces, molecules lose energy and combine to form complex organics.
6. These materials are later incorporated into forming planetary systems.
7. Stars and planets are born together as a natural consequence of cosmic processes
→ e.g., supernovae, galaxy collisions
6. The birth of planets and the formation of organic matter are natural phenomena
→ not mysteries, but outcomes governed by the laws of physics.

Formation of Molecules and Organics in Space



- In space, atoms cannot form molecules easily because there is no medium to absorb excess energy.
- One major scenario suggests that dust grains absorb this energy, allowing atoms to react slowly on their surfaces.
- Simple molecules (H₂, H₂O, CO) → complex organics (CH₃OH, even amino acids).
- A typical dust grain: dust core + ice mantle + organic layer

